Week 7 Problem Set: Rates and Tangents

Calculus Honors · Module 4: The Derivative · Lessons 4.01 and 4.02

Name: _____ Date: _____

> Show all of your work. A correct answer with no reasoning earns partial credit at most. Keep your writing neat and vertical, line up your equal signs, simplify the difference quotient fully before letting h shrink to 0, and label your final answers.

PART A · AVERAGE RATE OF CHANGE

1. For $f(x) = x^2 - 2x$, compute the average rate of change on $[1, 1+h]$ and simplify it to an expression in h. Then evaluate it at $h = 0.1$ and $h = 0.01$. What value does it approach as $h \to 0$?

2. A ball's height (in feet) t seconds after being thrown is $s(t) = -16t^2 + 40t + 5$. Find the average velocity over $[1, 1+h]$ as a simplified expression in h, then estimate the instantaneous velocity at $t = 1$ by evaluating at $h = 0.1$, $h = 0.01$, and $h = 0.001$.

PART B · THE DERIVATIVE AT A POINT

3. Use the definition $f'(a) = \lim_{h \to 0} \dfrac{f(a+h) - f(a)}{h}$ to find $f'(4)$ for $f(x) = x^2 - 3x$. Show the difference-quotient algebra in full.

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4. Find the equation of the tangent line to $f(x) = x^2 - 3x$ at $x = 4$, using your result from problem 3.

PART C · TANGENT LINES AND INTERPRETATION

5. The position of a car (in miles) after t hours on a highway is $p(t) = t^2 + 30t$. Find $p'(2)$ from the definition, and explain in one sentence what this number means about the car's motion at $t = 2$ hours, including the correct units.

PART D · EXPLAIN YOUR THINKING

6. Explain to a classmate, using the idea of a secant line becoming a tangent line, why shrinking h in the difference quotient toward 0 produces the exact instantaneous rate of change rather than just a good approximation of it. Use a specific example from this set as evidence.